THE BAND THEORY OF SOLIDS

* What are metals? Write some characteristic properties of metals.

Metals are naturally occuring chemical elements that are often hard, shiny, readily loses e^- to form +ve ions & can form a special metallic bond. Majority of the periodic table consists of metal. For example: iron, copper, zinc, aluminium, gold etc.

Metals are easily distinguishable by their properties. Some characteristic properties of metals are—

1. They have a metallic luster & can reflect light well. Hence they have a shiny appearance.
2. They are good conductors of heat & electricity.
3. Metals have high melting & boiling points.
4. Metals are usually dense.
5. Most metals are solid at room temperature; except mercury.
6. Metals are ductiles, i.e., metals can be drawn into wires without breaking, etc.

* As the temperature increases, the electrical resistance of a metal also increases - explain why.

The electrical resistance of a metal increases with increasing temperature. This is because when a metal is heated, the atoms in the metal vibrate with greater amplitude & frequency. Due to the increase in temperature, the thermal velocities of free e^- also increases. Therefore, the number of collisions bet^n free e^- & atoms increases. This phenomenon increases ~~the~~ opposition to the movement of e^-. Hence, the resistance increase with the increasing temperature.

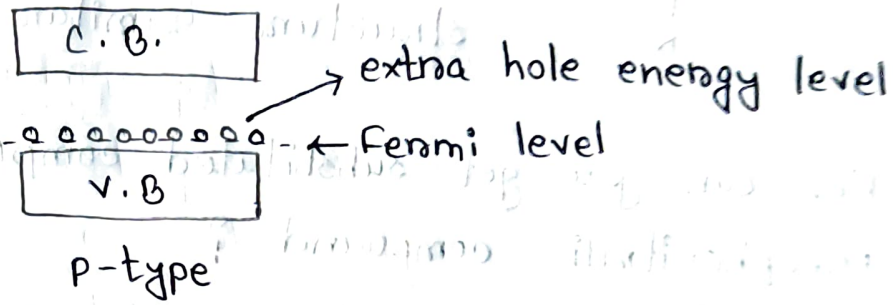
* What is p-type & n-type semiconductors?

Extrinsic or impure semiconductors can be classified depending upon the ~~im~~ type of impurity added in it. It is classified as -

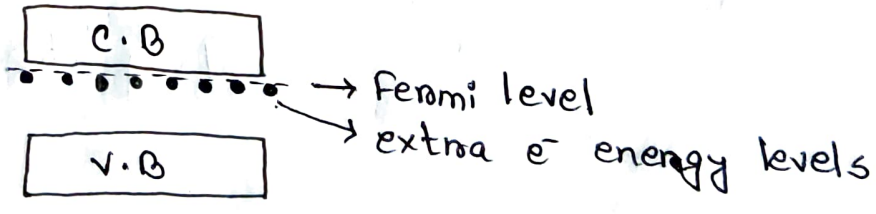
1. p-type semiconductors
2. n-type semiconductors

p-type : A p-type semiconductor is an intrinsic semiconductor doped with trivalent impurities, such as- boron, aluminium or gallium. In this semiconductor, the majority of charge carriers are holes which are

created from the deficiencies of valence e^- of the trivalent atom. The hole density of p-type semiconductor is much greater than the e^- density. The fermi level is closer to the valence band & away from the conduction band.



n-type : n-type semiconductors are intrinsic semiconductors doped with pentavalent impurities, such as antimony, arsenic or phosphorous. The majority of charge carriers are electrons here & hence the e^- density is much greater than the hole density. The fermi level in this lies closer to the conduction band & away from the valence band.



* Fermi Level: Fermi level is the highest energy level that an e^- can occupy at the absolute zero temperature.

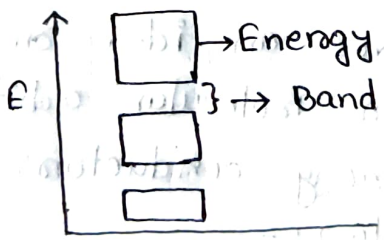
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* Discuss the main feature of Band Theory of Metals.

Band theory is a method to calculate the energy gap in electrons of a specific solid to separate the valence & conduction band.

When atoms come together to form a compound, their atomic orbitals combine together to form molecular orbitals with two distinct energy levels. As more atoms begin to mix & more molecular orbitals are formed, it is expected that many of these energy levels will start to be very close to each other & form energy continuum. These energy continuums are called bands, and this theory is known as the Band theory of solids.

In solid metals, the valence shells of each atom make contact & form molecular orbitals. When huge numbers of atoms combine to form a solid, the discrete energy levels are replaced by discrete ranges of energy or energy bands. It can be shown as —

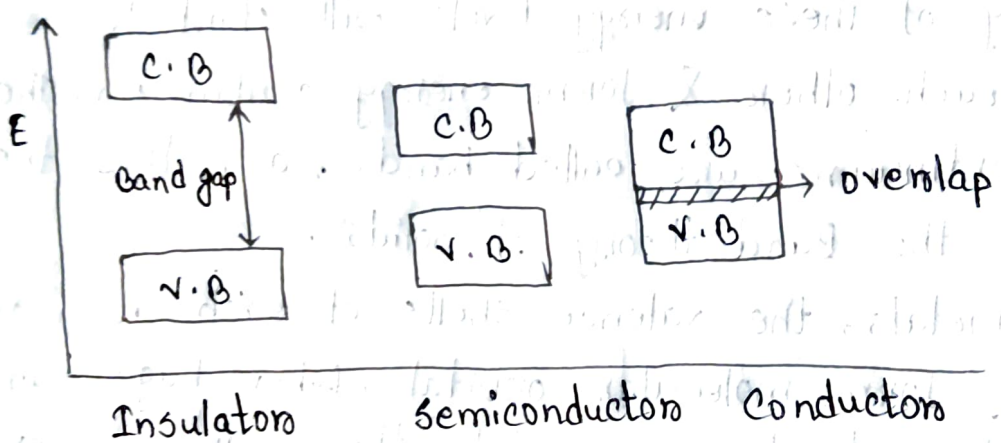


Metals can be classified based on this band theory as followed :—

1. Insulator : The solids in which the e^- cannot move from V.B to C.B because of the large gap betⁿ them is called insulators.

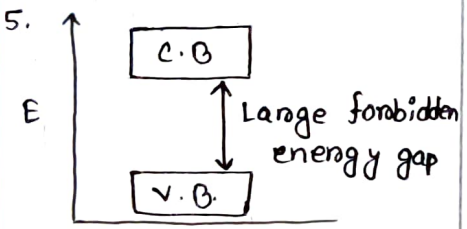
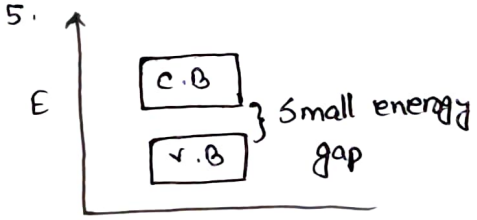
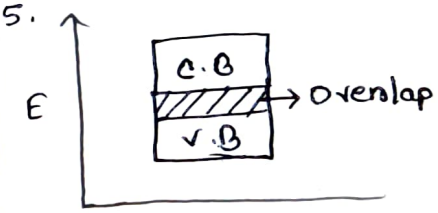
2. Conductor : The e^- can move easily from V.B to C.B in conductors as there is no band gap betⁿ them.

3. Semiconductor : Some e^- can leave the V.B for the C.B in semiconductor as the band gap betⁿ them is not so large.



* Show the differences among Conductors, Semiconductors & insulators on the basis of band theory of solids.

Band theory of solids gives an idea on how atomic orbita combine together to form molecular orbitals & energy bands. The differences among conductors, semiconductors & insulators is given below :—

Insulators	Semiconductors	Conductors
1. The e^- cannot move ea from V.B. to C.B.	1. Few e^- can move from V.B to C.B.	1. e^- can easily move from V.B to C.B.
2. Energy gap bet^n the bands is massive, almost ≥ 5 eV	2. Energy gap is not so large; about ≤ 3 eV.	2. No energy gap bet^n the bands; V.B & C.B. are overlapped.
~~3. The C.B is vacant~~	~~3. Conduction band is partially filled~~	~~3 C.B :~~
3. Does not conduct electricity.	3. Electricity conductivity is medium.	3. Can easily conduct electricity.
4. High resistivity	4. Medium resistivity	4. Very low resistivity.
5. 	5. 	5. 
6. Ex: Mica, Papers etc.	6. Ex: Si, Ge etc.	6. Ex: Cu, Al etc.

* **Alloy**: An alloy refers to a combination of two or more metals; or a metal combined with one or more elements. That is, an allow is a mixture of chemical elements of which at least one is a metal.

* **Doping**: Doping is the process of adding some impurity atom to a pure or intrinsic semiconductor to increase the conductivity of it.